

DOTE 6635: Artificial Intelligence for Business Research (Spring 2026)

Agentic AI

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Where Are We?

- Reasoning LLMs: Backbone for AI agents
- Replication automation, APE, Autoresearch, FARS: LLM Agents have driven paradigm shift in academic research.
- Agent-based modeling (e.g., structural models): An important and long-standing method in business, economics, and social science in general.
- AI agents are indeed impacting real business, from customer services, market research, coding co-pilot, to supply chain robotics.
- Automation ladder: Robustness (tedious digital labor) → collaboration (human interactions) → exploration (creativity, scientific discovery)

From ChatGPT to OpenClaw: AI that talks → AI that acts

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Agenda

- LLM Agents
- Skills
- OpenClaw
- Context Engineering

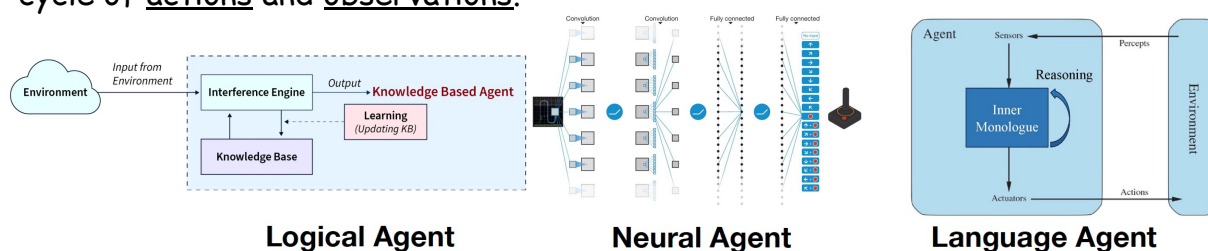
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What is Agent?

<https://language-agent-tutorial.github.io/slides/I-Introduction.pdf>; <https://llmagents-learning.org/slides/intro.pdf>

- "Definition": An intelligent system that interacts with some environment via a cycle of actions and observations.



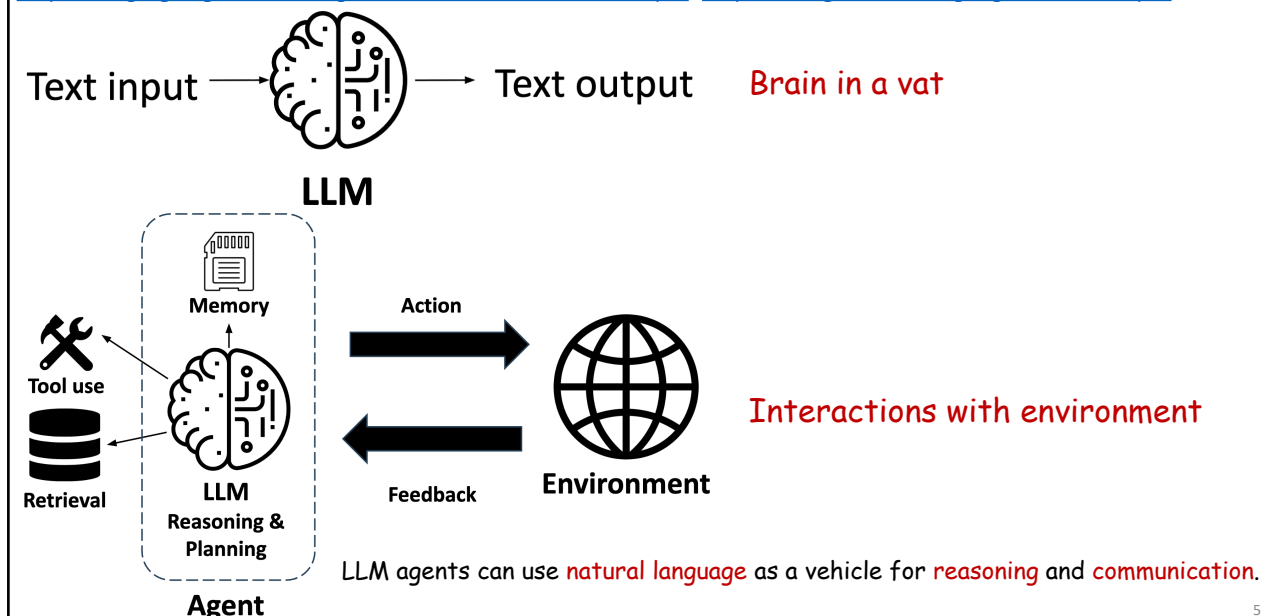
	Logical Agent	Neural Agent	Language Agent
Expressiveness	Low bounded by the logical language	Medium anything a (small) NN can encode	High almost anything, esp. verbalizable parts of the world
Reasoning	Logical inferences sound, explicit, rigid	Parametric inferences stochastic, implicit, rigid	Language-based inferences fuzzy, semi-explicit, flexible
Adaptivity	Low bounded by knowledge curation	Medium data-driven but sample inefficient	High strong prior from LLMs + language use

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From LLM to Agents

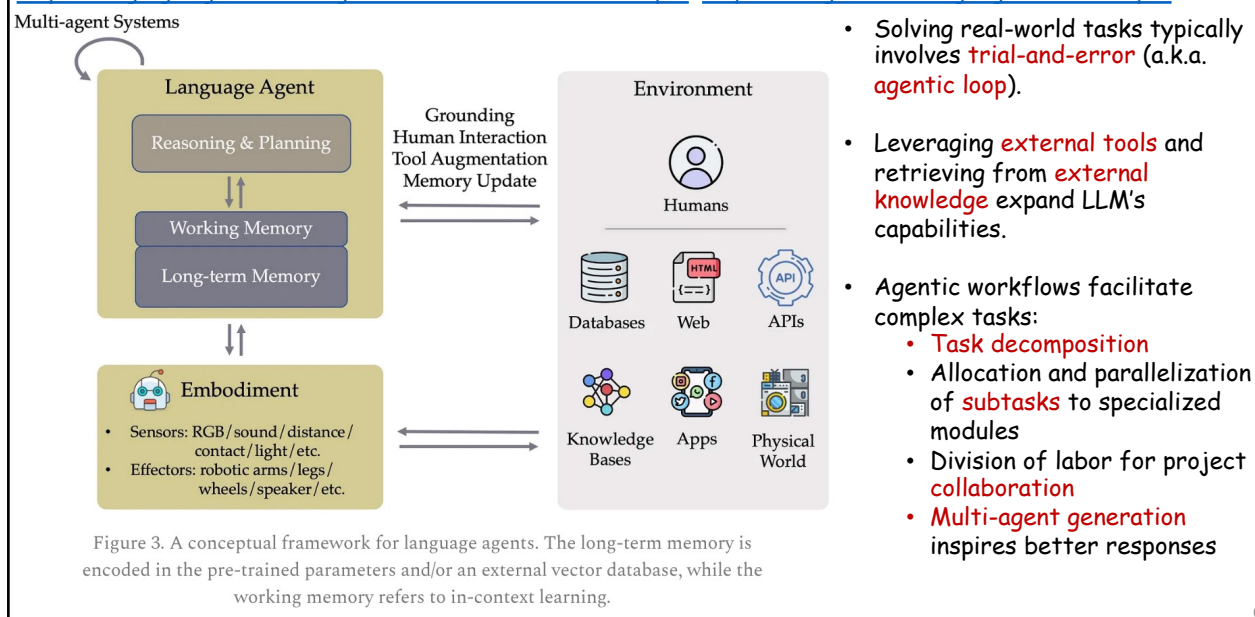
<https://language-agent-tutorial.github.io/slides/I-Introduction.pdf>; <https://llmagents-learning.org/slides/intro.pdf>



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Conceptual Framework of LLM Agents

<https://language-agent-tutorial.github.io/slides/I-Introduction.pdf>; <https://llmagents-learning.org/slides/intro.pdf>



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Agent Skills

<https://speech.ee.ntu.edu.tw/~hylee/ml/2026-course-data/intro.pdf>; <https://resources.anthropic.com/hubfs/The-Complete-Guide-to-Building-Skill-for-Claude.pdf>
<https://www.deeplearning.ai/short-courses/agent-skills-with-anthropic/>

- LLM agents are **smart** and **knowledgeable** PhDs
- Fundamentally, skills are **SOPs** for LLM agents to finish certain tasks.

3 Primary Skills Archetypes



Document & Asset Creation

Purpose: Consistent, high-quality outputs using built-in capabilities without external tools.

Key Techniques: Embedded style guides, structural templates, and quality checklists.

Real Example:
frontend-design
 (Generates production-grade UI components in React).



Workflow Automation

Purpose: Multi-step processes benefiting from a rigid, consistent methodology.

Key Techniques: Step-by-step validation gates, and iterative refinement loops.

Real Example:
skill-creator
 (Walks developers interactively through building new skills).



MCP Enhancement

Purpose: Workflow guidance that orchestrates and enhances tool access from an MCP server.

Key Techniques: Coordinating multiple sequential MCP calls, centralized error handling.

Real Example:
sentry-code-review
 (Analyzes and fixes GitHub PR bugs using Sentry data).

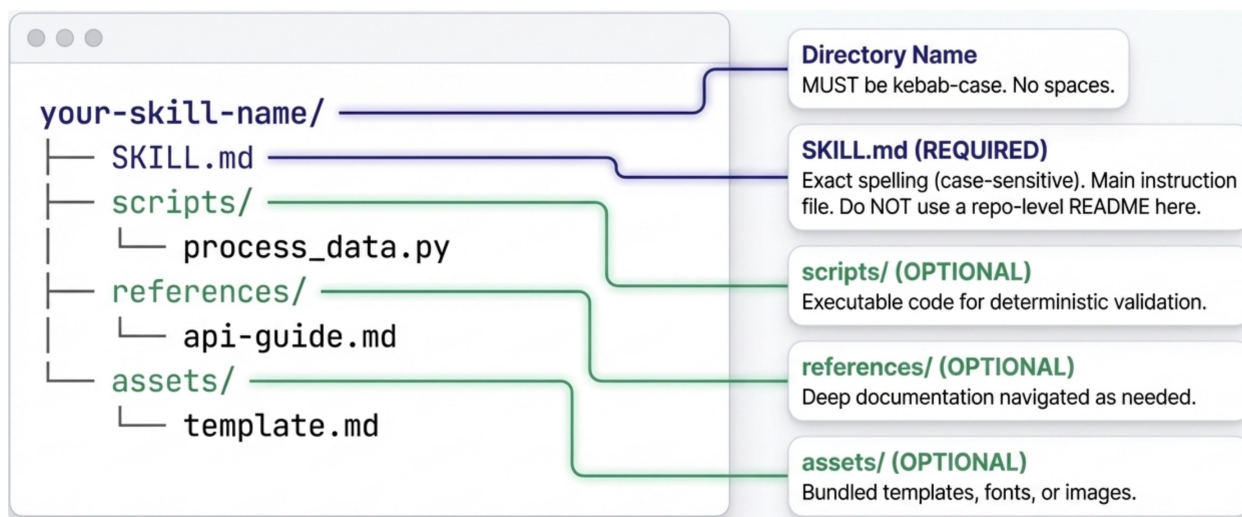
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Anatomy of Skills

<https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf>; <https://resources.anthropic.com/hubfs/The-Complete-Guide-to-Building-Skill-for-Claude.pdf>
<https://www.deeplearning.ai/short-courses/agent-skills-with-anthropic/>

- The file architecture may differ, but the logic remains the same for different AI agents.



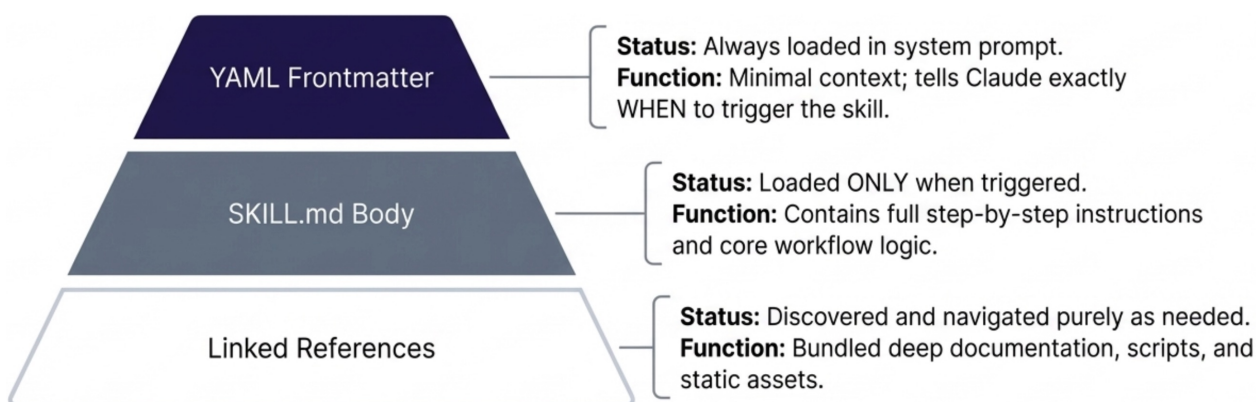
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Progressive Disclosure

<https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf>; <https://resources.anthropic.com/hubfs/The-Complete-Guide-to-Building-Skill-for-Claude.pdf>
<https://www.deeplearning.ai/short-courses/agent-skills-with-anthropic/>

- LLM agents should access deep knowledge without easily exhausting context windows.
- Progressive disclosure **minimizes token usage** while maintaining access to **specialized expertise**.



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Frontmatter: When the Skill is Triggered

<https://speech.ee.ntu.edu.tw/~hylee/ml/2026-course-data/intro.pdf>; <https://resources.anthropic.com/hubfs/The-Complete-Guide-to-Building-Skill-for-Claude.pdf>
<https://www.deeplearning.ai/short-courses/agent-skills-with-anthropic/>

- This metadata is loaded into the **system prompt** and determines **if your skill is triggered**.

Requires standard
YAML delimiters.

```

---
name: sprint-planner
description: Manages Linear project workflows. Use
            when user mentions 'sprint', 'Linear tasks', or
            asks to 'create tickets'.
metadata:
  version: 1.0.0
            
```

CRITICAL: kebab-case
only. No spaces, no
caps. Cannot contain
'claude' or 'anthropic'.

The Trigger Formula: [What it does]
+ [When to use it / Trigger phrases].
Must be under 1024 chars.

⚠️ **SECURITY RESTRICTION:** No XML
angle brackets (< >) are permitted
anywhere in the frontmatter.

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Nonambiguous Instructions

<https://speech.ee.ntu.edu.tw/~hylee/ml/2026-course-data/intro.pdf>; <https://resources.anthropic.com/hubfs/The-Complete-Guide-to-Building-Skill-for-Claude.pdf>
<https://www.deeplearning.ai/short-courses/agent-skills-with-anthropic/>

- Code is deterministic, but natural language is not. Write instructions that leave no room for guesswork.

● ● ●
DON'T: Vague & Ambiguous
✕

Validate the data before proceeding. Make sure to handle errors.

Why it fails:

- Relies entirely on LLM interpretation of 'valid'.
- Ambiguous language.
- Lacks concrete error recovery paths.

● ● ●
DO: Specific & Actionable
✓

CRITICAL: Before calling create_project, verify:

- Project name is non-empty

Run python scripts/validate.py. If validation fails, check:

- Invalid date formats (use YYYY-MM-DD)

Why it succeeds:

- Explicit numbered logic.
- Defers to deterministic Python scripts for validation.
- Includes exact error recovery instructions.

- You can download useful skills from <https://github.com/anthropics/skills> or other skills repo/hub and **ask your AI agents to write skills** for you based on the best practices <https://agentskills.io/home>.

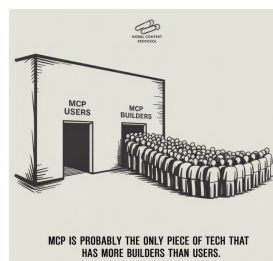
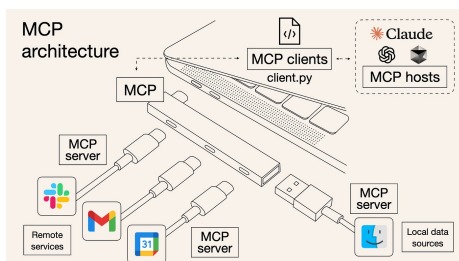
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Skills & MCPs

<https://speech.ee.ntu.edu.tw/~hylee/ml/2026-course-data/intro.pdf>; <https://resources.anthropic.com/hubfs/The-Complete-Guide-to-Building-Skill-for-Claude.pdf>
<https://www.deeplearning.ai/short-courses/agent-skills-with-anthropic/>; <https://www.anthropic.com/news/model-context-protocol>

- Model context protocol (MCP) is a standard for connecting AI agents to apps, data and tools.



MCP (Connectivity)	Skills (Knowledge)
Connects Claude to your service (Notion, Asana, Linear, etc.)	Teaches Claude how to use your service effectively
Provides real-time data access and tool invocation	Captures workflows and best practices
What Claude can do	How Claude should do it

The kitchen analogy

MCP provides the professional kitchen: access to tools, ingredients, and equipment.

Skills provide the recipes: step-by-step instructions on how to create something valuable.

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OpenClaw



<https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf>; <https://voge.ai/openclaw-en.html>; <https://openclaw.ai/>; <https://github.com/openclaw/openclaw>; <https://clawhub.ai/>

- OpenClaw (formerly Clawbot, Moltbot, and Molty) is an open-source agentic system that executes tasks via LLMs using messaging platforms (e.g., WhatsApp, Lark, Discord) as its main user interface.

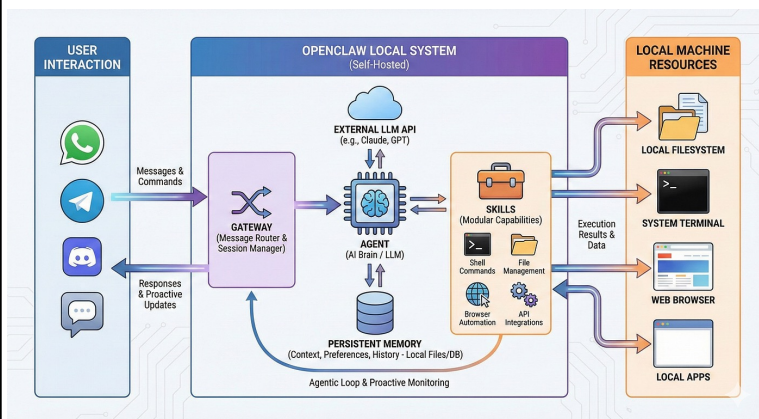
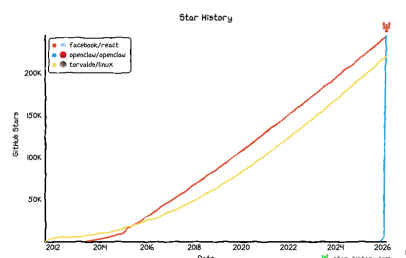
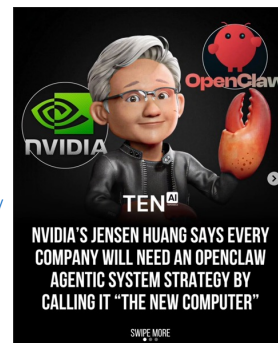


Illustration of OpenClaw Architecture

OpenClaw and Token Economy

<https://www.nvidia.com/get/keynote/>



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Behind OpenClaw's Viral Growth



<https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf>; <https://voge.ai/openclaw-en.html>; <https://openclaw.ai/>; <https://github.com/openclaw/openclaw>; <https://clawhub.ai/>

	Consumer Chat (ChatGPT)	Pro Local (Cursor/Codex)	Viral Agentic (OpenClaw)
Capability	Pure Chat & Reasoning	Agentic Loop / File I/O	Agentic Loop / File I/O
Interface	Web UI	IDE / Terminal	WhatsApp, Slack, Lark
Accessibility	Mainstream	Niche (Programmers)	Mainstream

Just as DeepSeek democratized reasoning, OpenClaw democratized the local programming agent by stripping away the IDE and placing it in familiar chat apps.

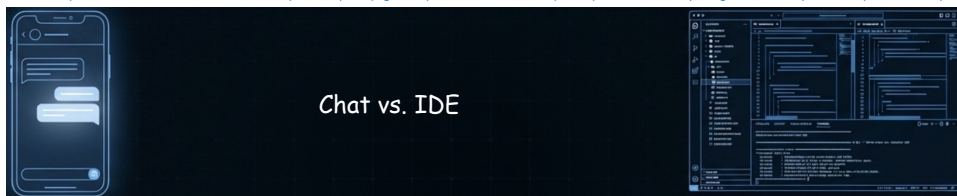
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Approachability vs. Deep Work



<https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf>; <https://vage.ai/openclaw-en.html>; <https://openclaw.ai/>; <https://github.com/openclaw/openclaw>; <https://clawhub.ai/>



Dimension	Chat-Based AI (OpenClaw)	Workspace AI (Cursor/OpenCode)
Workflow Path	Linear & Clunky →	Multi-threaded & Branching 
Information Density	Low	High
Observability	Poor (Black Box)	Transparent (Real-time tracking)
Target Audience	Broad Consumer	Deep Knowledge Worker

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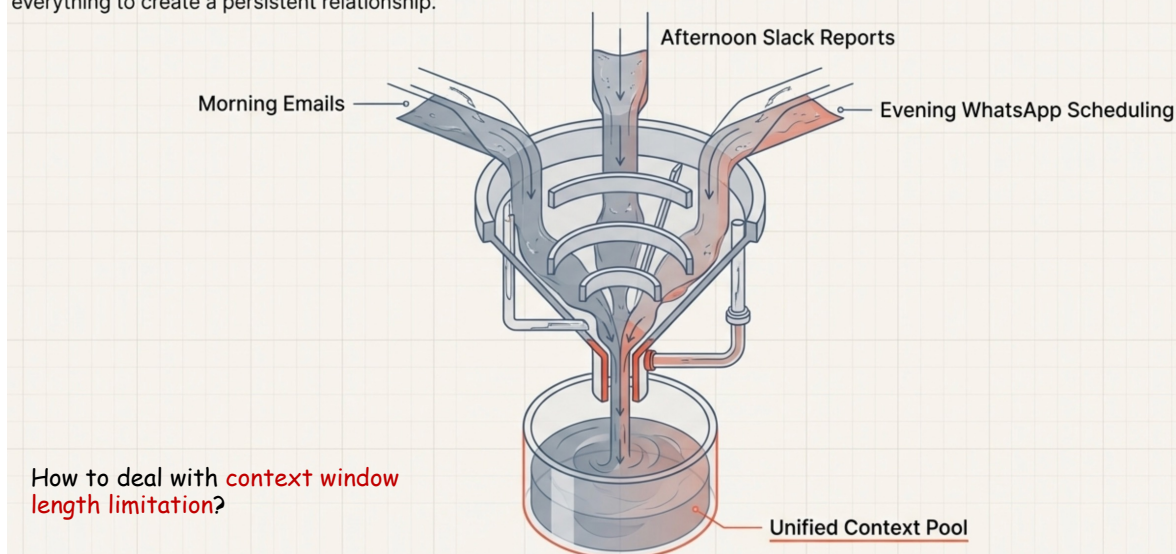
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Unified Context



<https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf>; <https://vage.ai/openclaw-en.html>; <https://openclaw.ai/>; <https://github.com/openclaw/openclaw>; <https://clawhub.ai/>

Pro tools isolate context. OpenClaw defaults to mixing everything to create a persistent relationship.



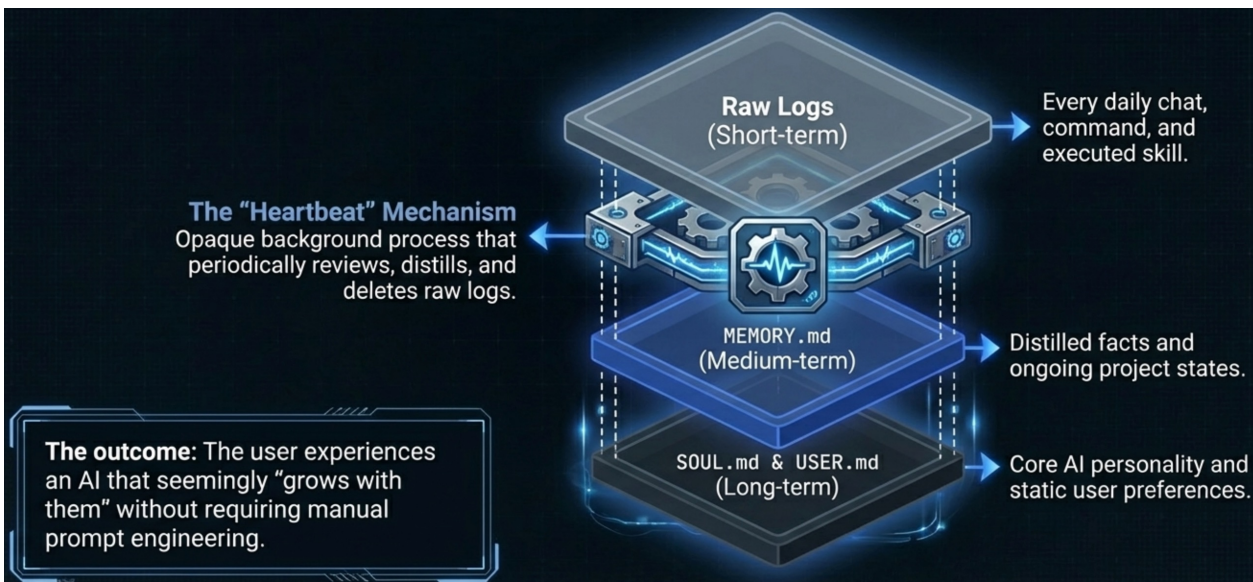
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Self-Evolving Memory Engine



<https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf>; <https://voge.ai/openclaw-en.html>; <https://openclaw.ai/>; <https://github.com/openclaw/openclaw>; <https://clawhub.ai/>



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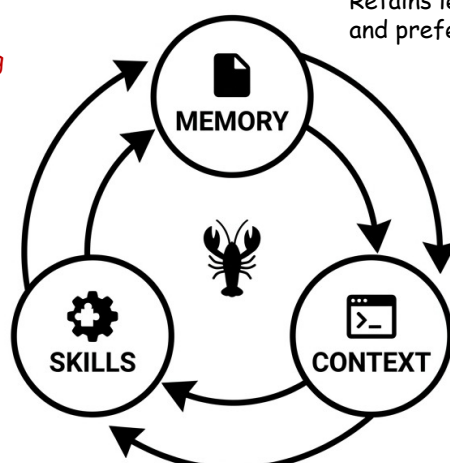
OpenClaw Flywheel



<https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf>; <https://voge.ai/openclaw-en.html>; <https://openclaw.ai/>; <https://github.com/openclaw/openclaw>; <https://clawhub.ai/>

These 3 designs are **reinforcing each other**, achieving **continual learning**.

Ecosystem of Skills:
Emergent business capabilities and the ability to self-evolve by writing its own code.



Persistent Memory:
Retains learnings, habits and preferences over time.

Unified Context Pool:
Mixes inputs from multiple platforms (WhatsApp, Telegram, Discord, etc.) into a single data stream.

The AI that actually does things.

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Blackboxed Memory



<https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf>; <https://yoge.ai/openclaw-en.html>; <https://openclaw.ai/>; <https://github.com/openclaw/openclaw>; <https://clawhub.ai/>

The OpenClaw Reality

Unpredictable Purges: The background heartbeat might automatically summarize or delete a 5,000-word research report.

Cross-Context Interference: A temporary preference from Project A mysteriously pollutes Project B.

Opaque Logic: Zero visibility into why the AI merged or dropped specific thoughts.

The Pro Requirement

Long-term Asset Building: Knowledge must be a versioned, explicitly managed asset.

Absolute Sources of Truth: Files like docs/research.md must be strictly referenced, never auto-summarized by a background process.

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Lethal Trifecta of Agentic AI



<https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf>; <https://yoge.ai/openclaw-en.html>; <https://openclaw.ai/>; <https://github.com/openclaw/openclaw>; <https://clawhub.ai/>

- The \$16M \$CLAWD Token Scam



- 12% of Skills Contain Malicious Code

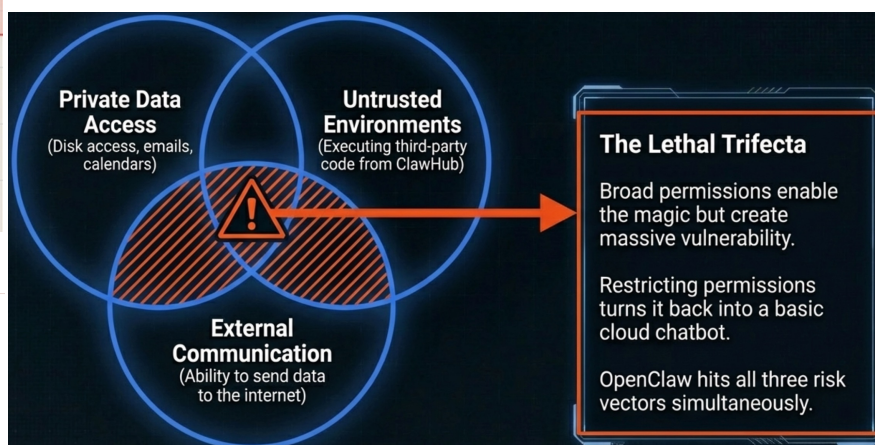
ERROR

KDI RESEARCH

ClawHavoc: 341 Malicious Clawed Skills Found by the Bot They Were Targeting



Alex, Oren Yomtov
February 1, 2026



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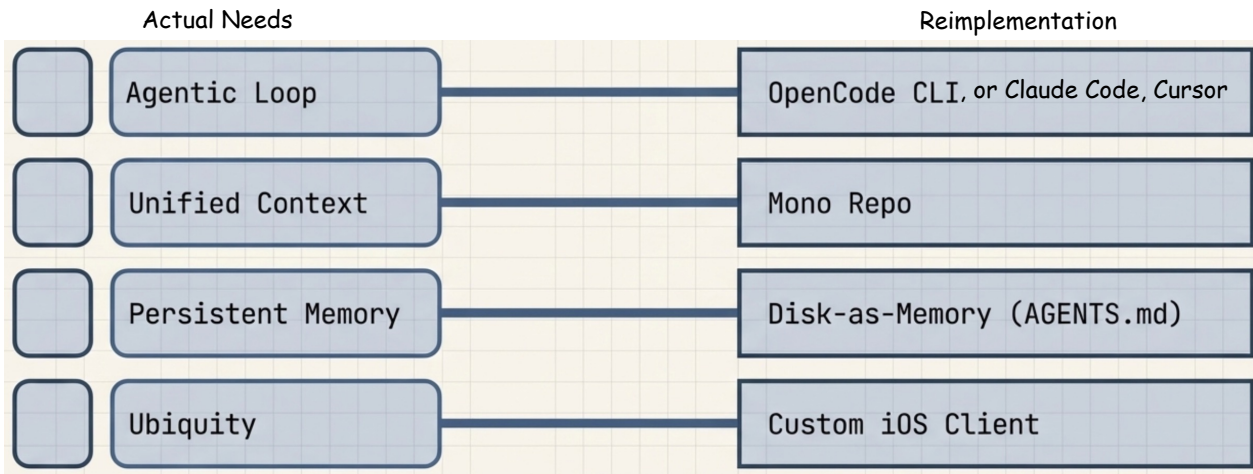
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Reimplementation Core Architecture



<https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf>; <https://vage.ai/openclaw-en.html>; <https://openclaw.ai/>; <https://github.com/openclaw/openclaw>; <https://clawhub.ai/>

OpenClaw's viralness demands **massive compromises**, but we can reconstruct the magic within a **controllable architecture**.



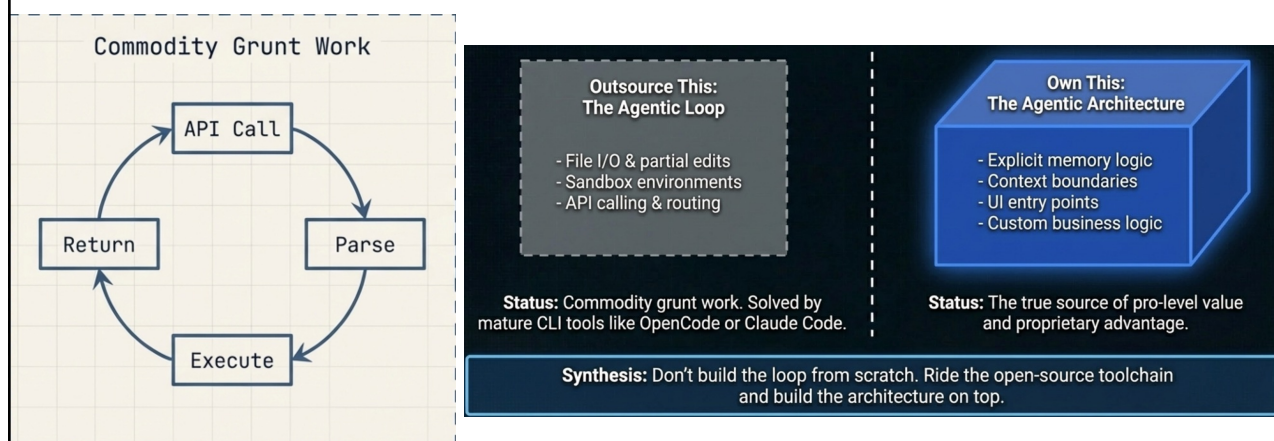
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Outsourcing Agentic Loops



<https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/intro.pdf>; <https://vage.ai/openclaw-en.html>; <https://openclaw.ai/>; <https://github.com/openclaw/openclaw>; <https://clawhub.ai/>



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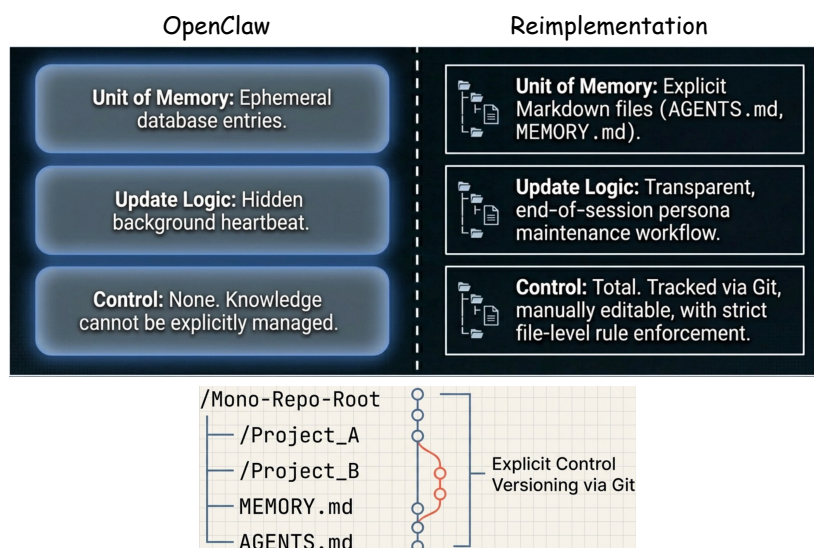
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Redesign Memory Architecture



<https://speech.ee.ntu.edu.tw/~hylee/ml/2026-course-data/intro.pdf> <https://voge.ai/openclaw-en.html> <https://openclaw.ai/> <https://github.com/openclaw/openclaw> <https://clawhub.ai/>

Restore **fine-grained control over knowledge assets and context** using the file system (disk-as-memory) and Git (mono repo).



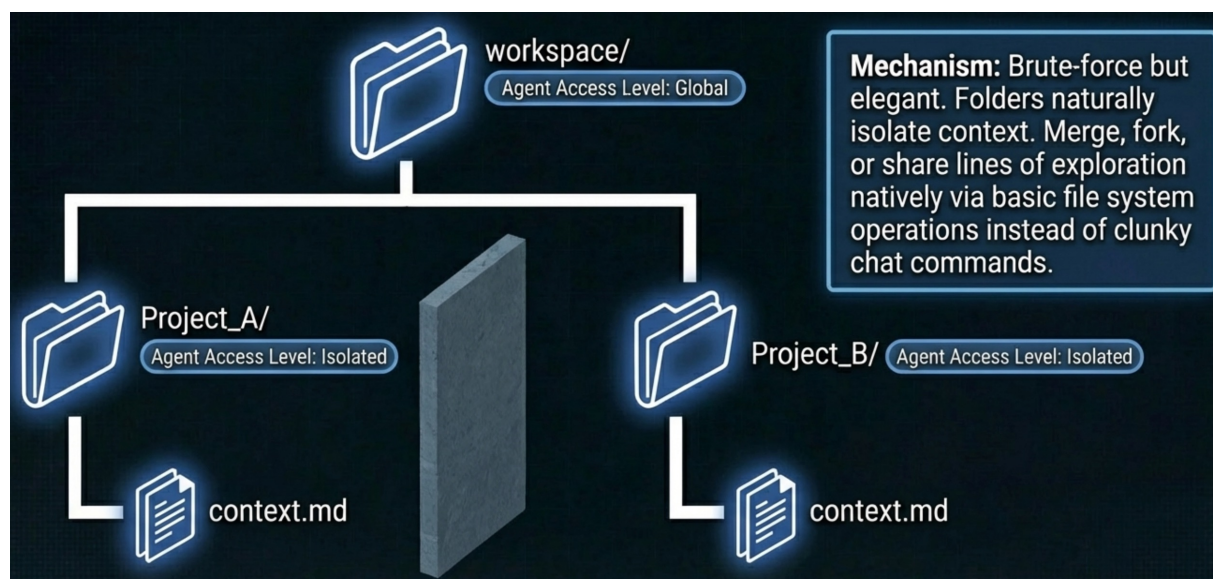
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Mono-repo Solves Context Pollution



<https://speech.ee.ntu.edu.tw/~hylee/ml/2026-course-data/intro.pdf> <https://voge.ai/openclaw-en.html> <https://openclaw.ai/> <https://github.com/openclaw/openclaw> <https://clawhub.ai/>



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One More Thing: Secure the Skills Supply Chain

<https://speech.ee.ntu.edu.tw/~hy/ee/ml/2026-course-data/intro.pdf>; <https://voge.ai/openclaw-en.html>; <https://openclaw.ai/>; <https://github.com/openclaw/openclaw>; <https://clawhub.ai/>

Blindly installing 3rd-party MCP servers risks **catastrophic supply chain attacks**.



It takes only **minutes** in the age of agentic AI coding, reducing **execution risk to zero 0**.

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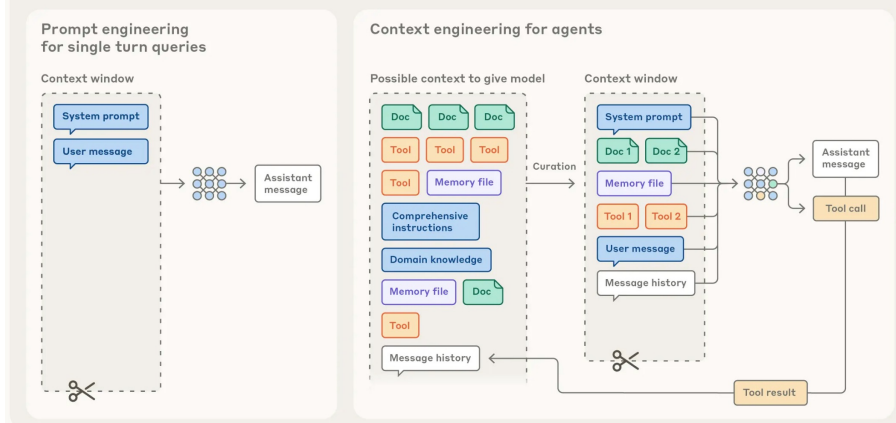
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Context is the Key

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; <https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents>

- **Context engineering:** Curating and maintaining the optimal set of information during LLM inference.

Prompt engineering vs. context engineering



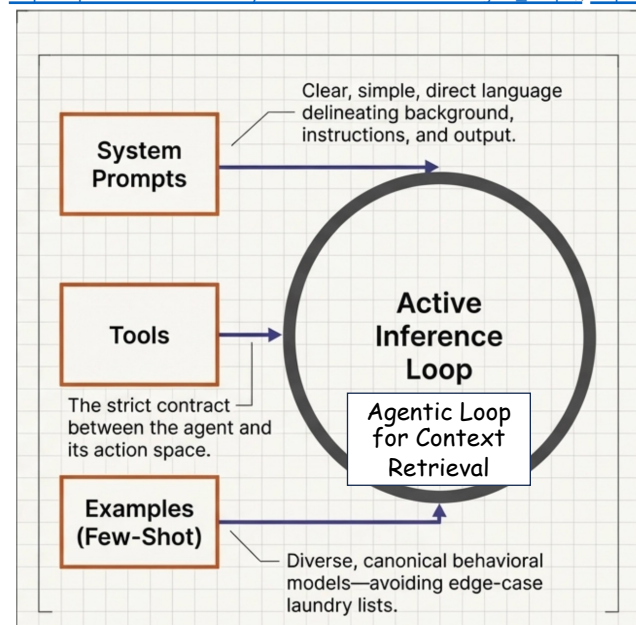
In contrast to the discrete task of writing a prompt, context engineering is iterative and the curation phase happens each time we decide what to pass to the model.

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Anatomy of Effective Context

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; <https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents>



- Anthropic's definition of agents: LLMs **autonomously** using tools in a **loop**.
 - Smarter models allow agents to **more independently** navigate nuanced problem spaces and recover from errors.
- **System prompts** define the agent **attitude**.
- **Tools** define the strict **contract** between agents and their **information/action space**.
- Provide **examples** as few-shot in-context learning.

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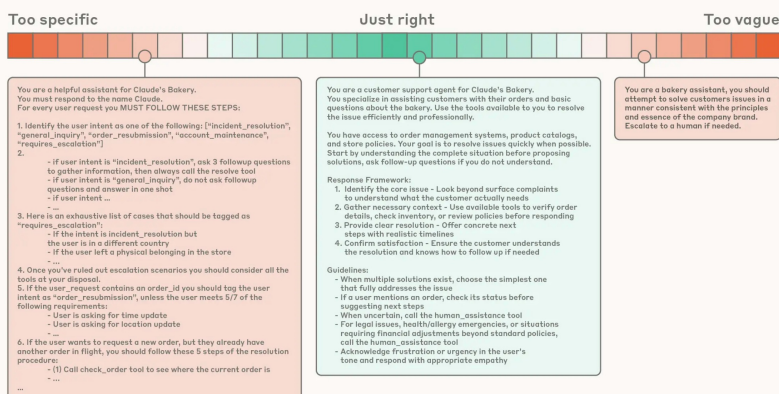
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System Prompts

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; <https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents>

- **System prompts:** The prompt that defines the **right altitude** for the agent.

Calibrating the system prompt



Organize the prompts into **distinct sections** like

- **<background_info>**
- **<instructions>**
- **### Tool guidance**
- **## Output description**

At one end of the spectrum, we see brittle if-else hardcoded prompts, and at the other end we see prompts that are overly general or falsely assume shared context.

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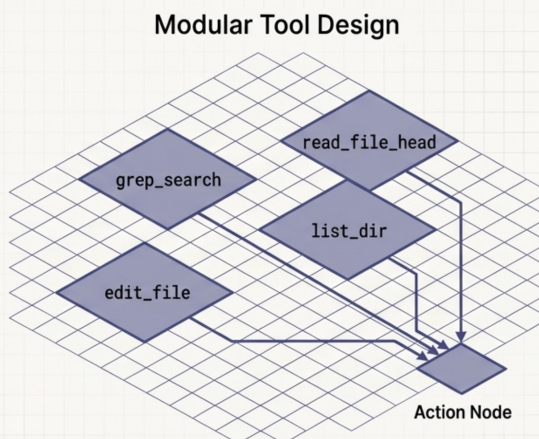
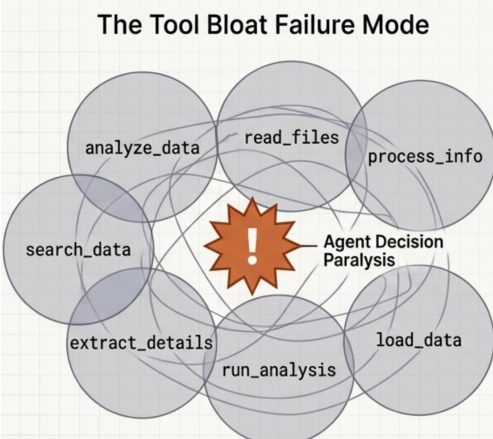
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Token-Efficient Action Space

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; <https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents>

Defining a Token-Efficient Action Space

Tools define the strict contract between agents and their environment. A bloated toolset with overlapping functionality guarantees ambiguity and wastes tokens. If a human engineer cannot definitively choose which tool to use in a scenario, an AI agent will also freeze or hallucinate.



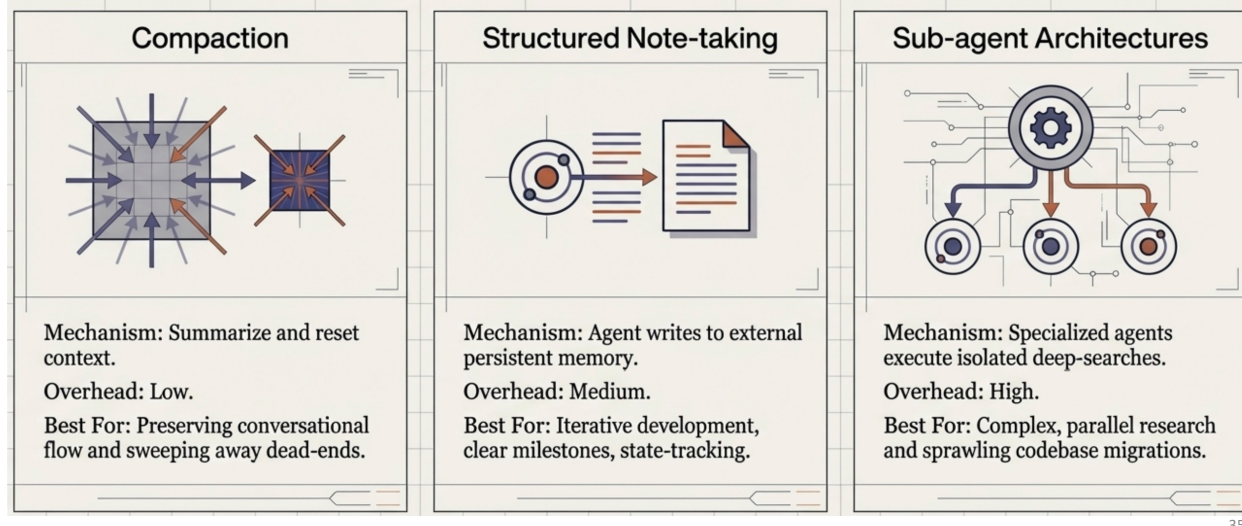
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Architecting for Long Horizons

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; <https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents>

For continuous tasks spanning tens of minutes to multiple hours, simply waiting for larger context windows fails due to inevitable context pollution. Sustaining coherence requires architectural interventions that actively prune and persist state outside the primary context window.

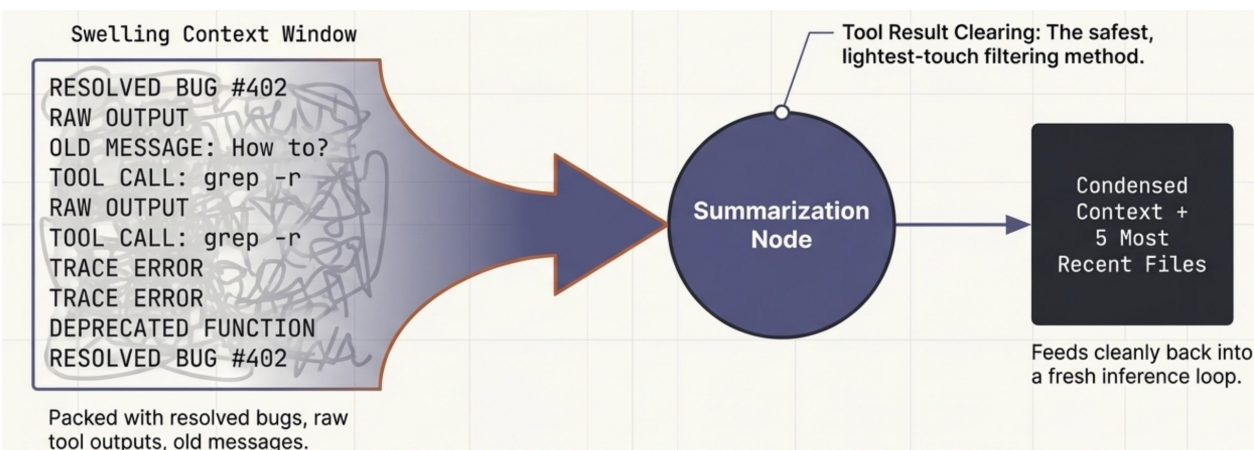


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Compaction: Distilling the State

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; <https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents>



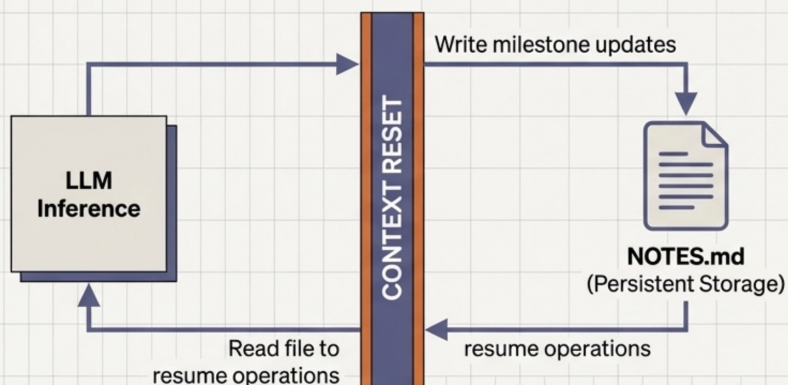
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Agentic Memory: Structured Note-Taking

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; <https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents>

By equipping agents with tools to read and write to persistent file systems, they can track complex dependencies across thousands of steps. This simple architectural pattern acts as persistent memory, allowing an agent to maintain strategic coherence across massive context resets.



The Pokémon Example

Target: Train to Level 10.

Current state: Route 1.

Tally: Pikachu +8 levels.

Strategy: Use Thunder Shock on Pidgey.

State tracked entirely outside the active context window.

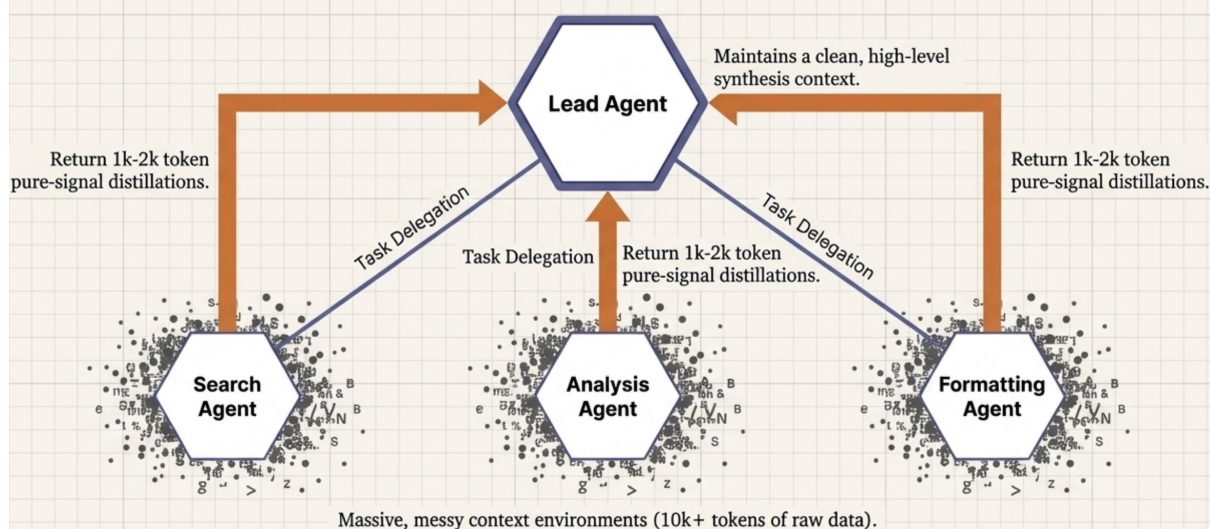
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Sub-agent Architectures

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; <https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents>

Rather than forcing one agent to maintain state across an entire project, isolate context pollution. Sub-agents execute deep technical exploration in dirty, high-token environments, but return only highly condensed, pure-signal summaries to the orchestrating lead agent.



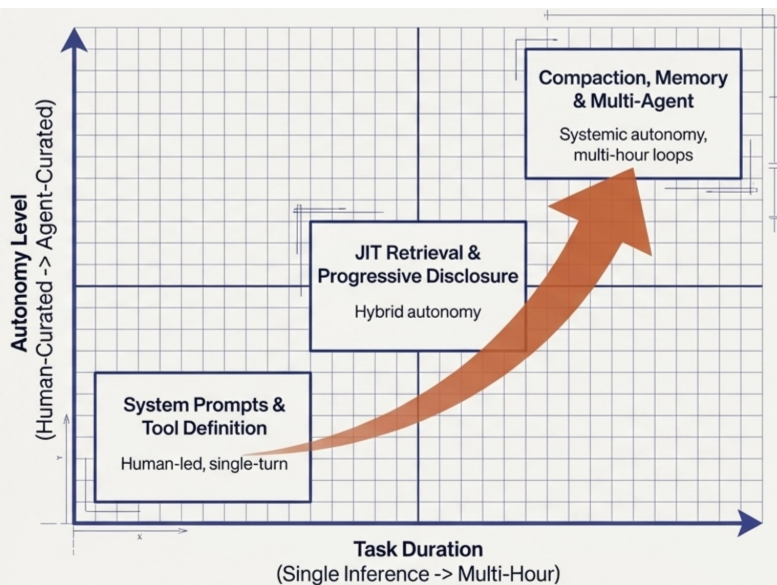
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Context Engineering Capability Map

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; <https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents>

The trajectory of AI engineering is clear. As time horizons expand and reasoning capabilities scale, human-curated context must yield to systemic architectures. The ultimate goal is building environments where the LLM dynamically curates its own attention budget.



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Context is a Precious Resource

As underlying models grow smarter, they require less prescriptive hand-holding and demand more architectural freedom. Whether building lightweight single-turn tools or orchestrating multi-hour autonomous clusters, the mandate remains unchanged: engineer the environment to deliver the absolute highest signal with the minimal required tokens.

https://speech.ee.ntu.edu.tw/~hylee/ml/ml2026-course-data/agent_era.pdf; <https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents> 40

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Context Engineering

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- **Context engineering** can be represented in the following abstract format:

$I_1 \leftarrow$ initial input

$C_1 \leftarrow$ empty

For $t = 1$ to ∞

$$O_t = LLM(I_t, C_t)$$

$$C_{t+1} \leftarrow F(C_t, I_t, O_t)$$

Context window
explodes easily.

$I_1 \leftarrow$ initial input

$C_1 \leftarrow$ empty

For $t = 1$ to ∞

$$O_t = LLM(I_t, P_t)$$

$$C_{t+1} \leftarrow F(C_t, I_t, O_t)$$

$$\{P_{t+1}, M_{t+1}\} \quad \{P_t, M_t\}$$

$$C = \{P, M\}$$

P_t is the part of
information delivered
to the context of LLM.

How should we properly design P_t ?

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